Components of Buildings contd…..

FOUNDATION

- Foundation is that part of the substructure of the building which safely transmits the load of the superstructure to the soil beneath it and the sides of the wall.
- The bearing capacity of soil is defined as the maximum load per unit area, which the soil can support without yielding.
- In designing the foundation, the safe bearing capacity of the soil which is the ultimate bearing capacity reduced by a factor of safety should be considered.
-

Types of Foundations

- Foundations can be broadly divided into two categories

1. Shallow foundations and

2. Deep foundations.

- Generally if the depth to width ratio of the foundation is less than 1, then it is known as a shallow foundation and if it is greater than 1, it is known as deep foundation.

Shallow Foundation

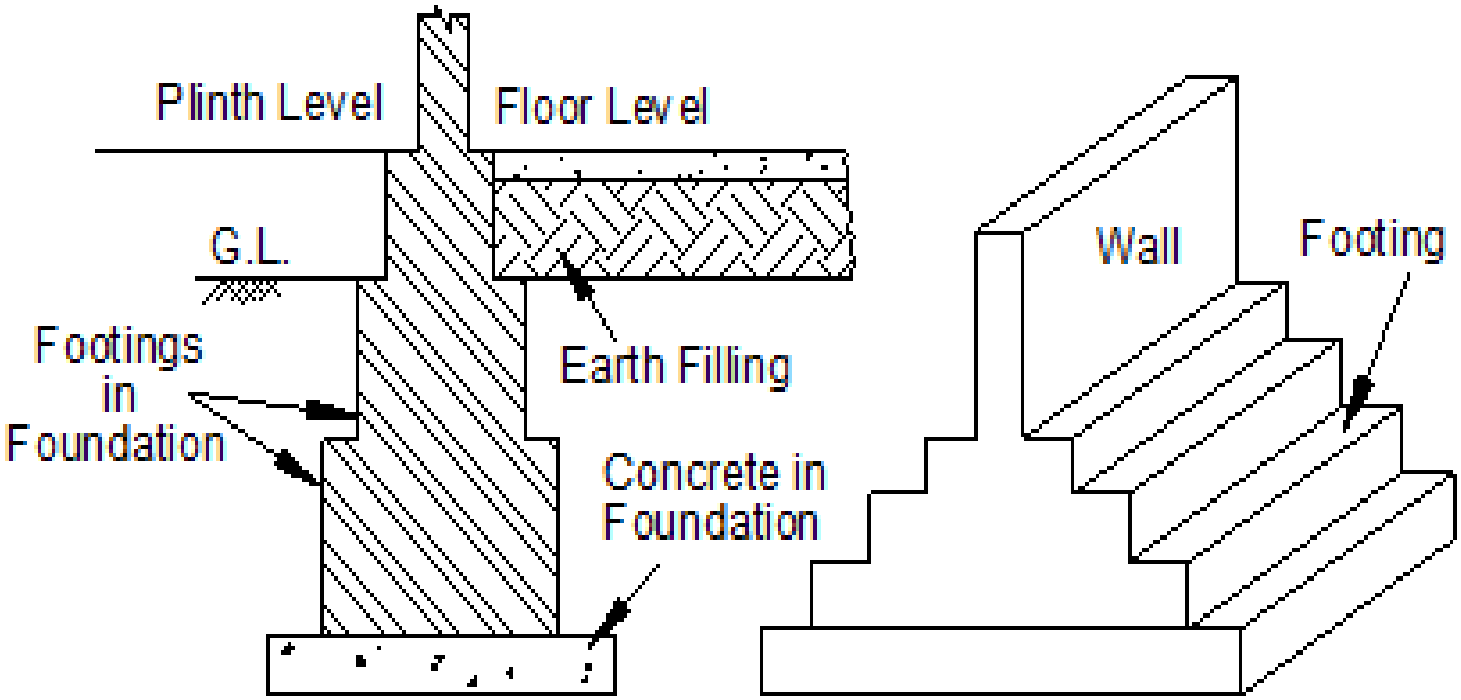
The various types of shallow foundations are as follows:

- Wall footing
- Isolated footing
- Combined footing
- Inverted arch footing
- Continuous footing
- Cantilever footing
- Grillage foundation
- Raft or Mat foundation

1.Wall Footing

This footing is provided throughout the length of the wall and may be either stepped or single. These are used for light structures.

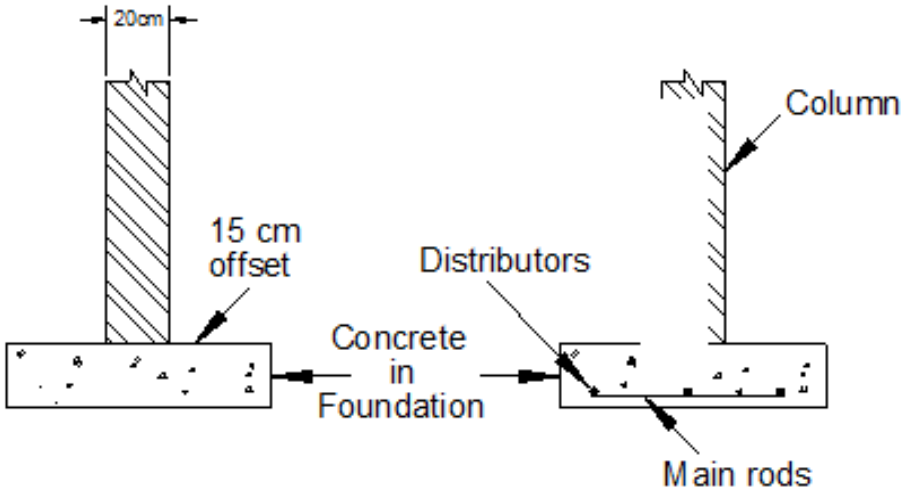
E.g. Load bearing walls.

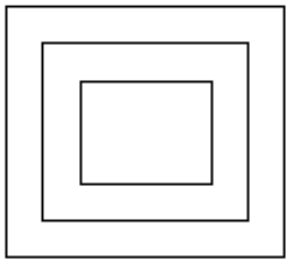




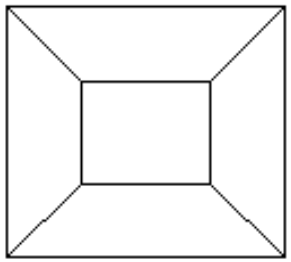
2. Isolated or Column Footing

This footing supports individual columns. It may be simple, stepped or sloped footing.





Stepped Footing



Sloped Footing

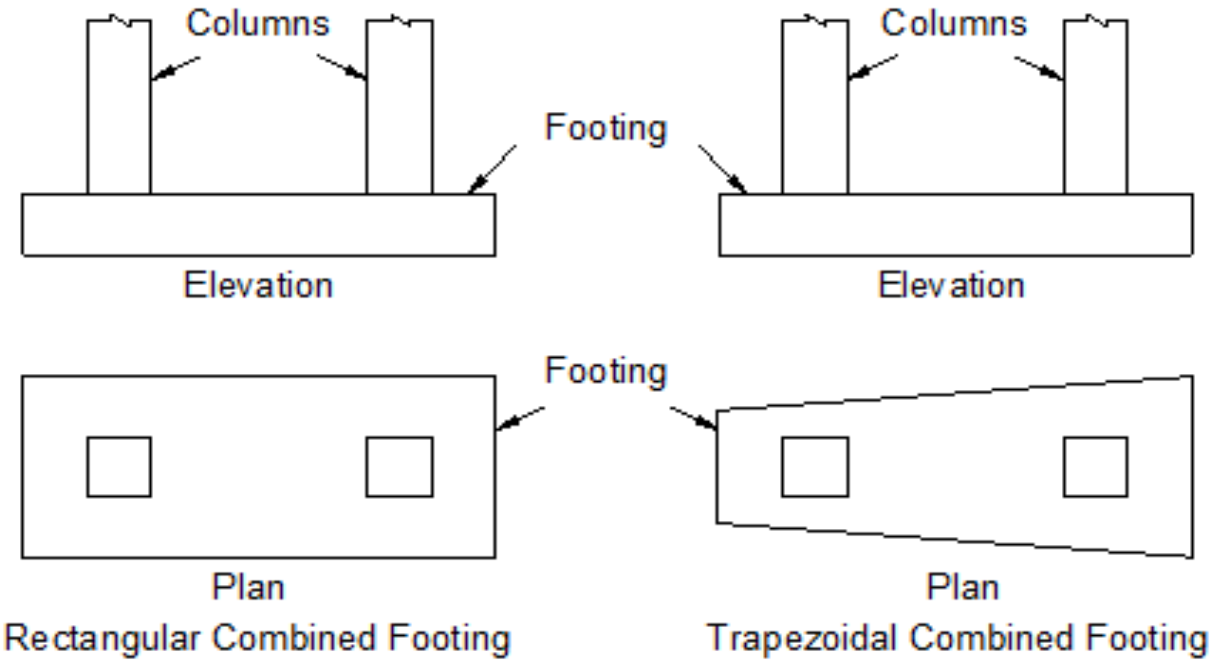


3. Combined Footing

- If the columns carry equal loads, then a rectangular shaped footing can be provided.
- If there are unequal loads then trapezoidal footing can be constructed.
- the center of gravity of the resultant area should coincides with the center of gravity of the loads.



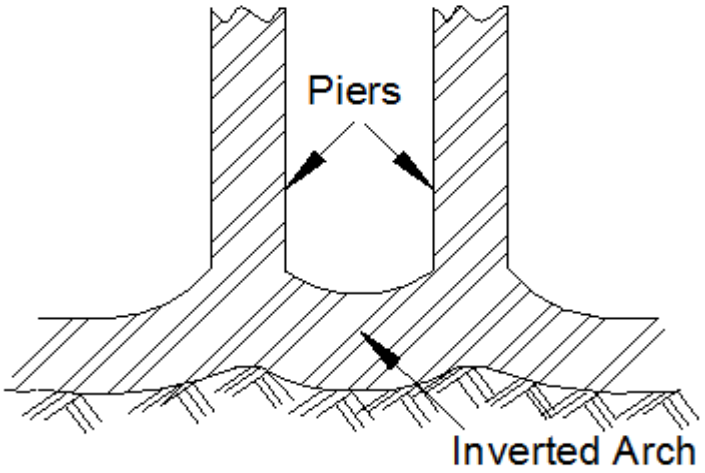
Combined Footing contd.....



4. Inverted Arch Footing

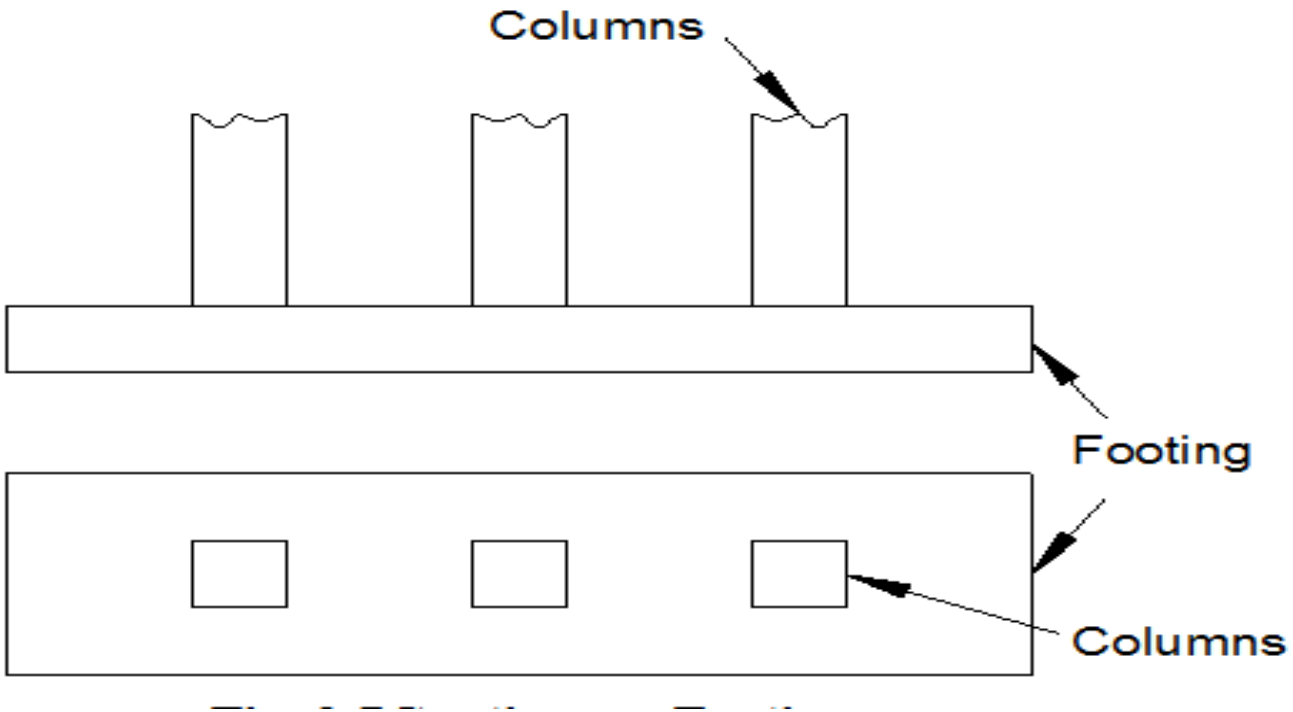
- It is constructed on soft soils.

- This type of foundation is suitable for bridges, reservoirs and tanks.



5. Continuous Footing

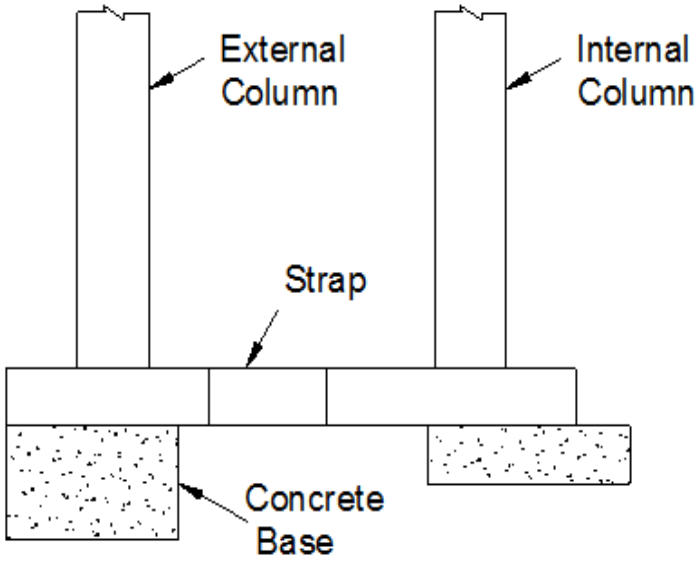
- This consists of a single continuous reinforced concrete slab supporting three or more columns in a row.
- This prevents differential settlement in the structure.





6. Strap or Cantilever Footing

- footing connected by a beam called as strap.
- Provided when the distance between the columns is great.
- The strap beam does not have any contact with the soil.
- The column loads are transferred into the soil uniformly.



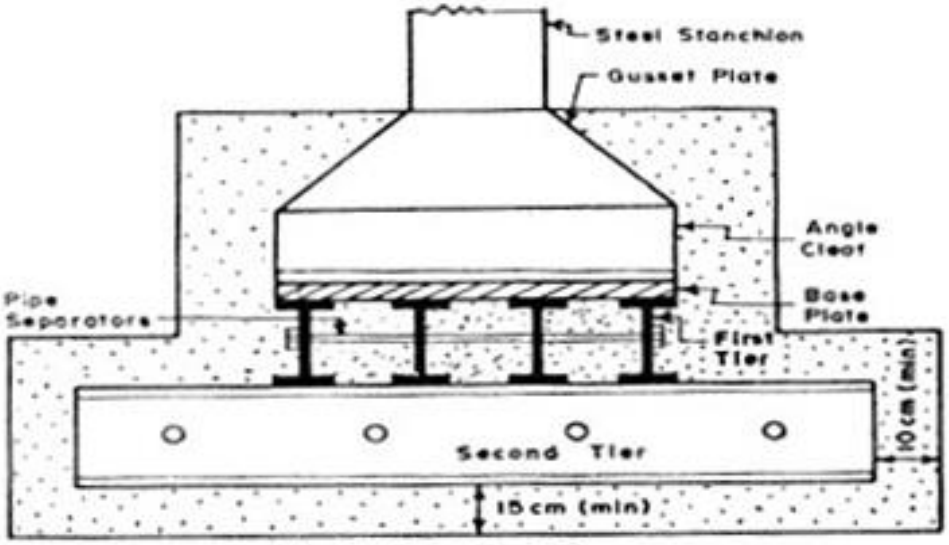
7. Grillage Foundation

- The grillage footing transfers heavy loads from steel columns to the foundation.

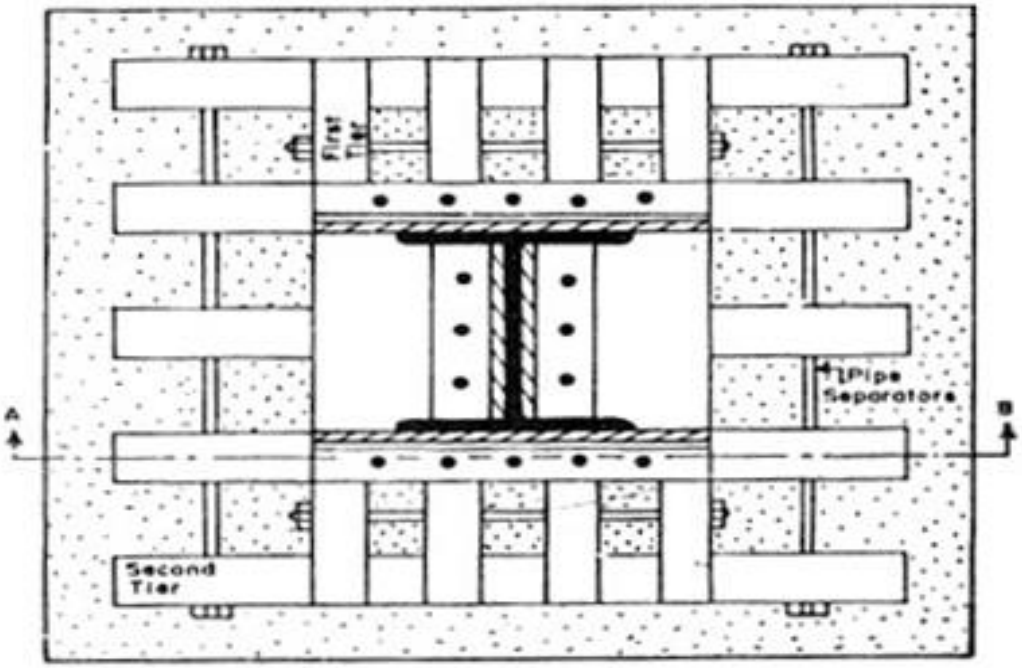
- The steel joints are held in place by pipe separators and nuts are completely embedded in concrete of minimum 15 cm to prevent corrosion.

- This type of foundation can be constructed for a single column or more than one column.





(a) Section of AB

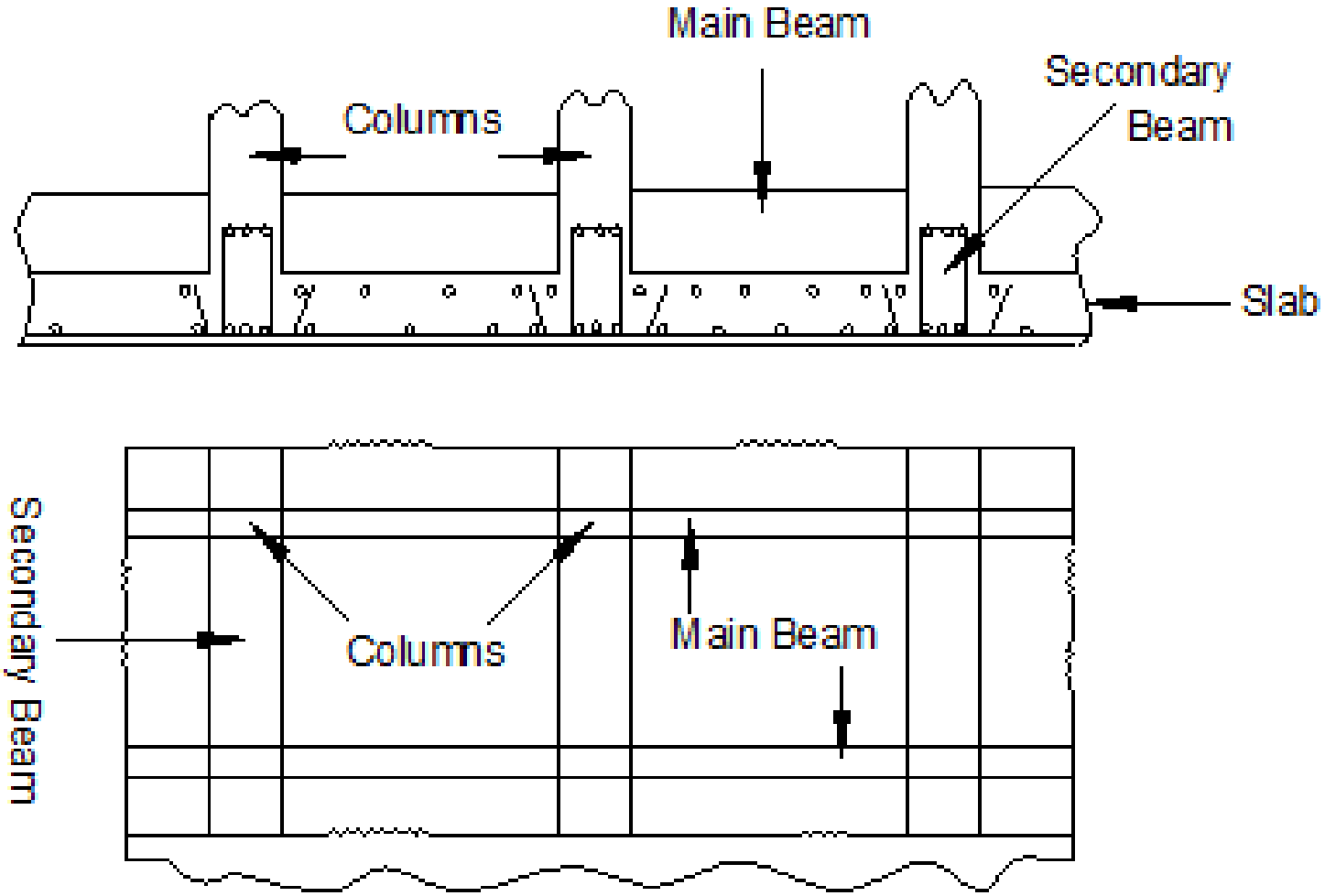


(b) Plan

8. Raft or Mat Foundation

- where the allowable bearing capacity is low and the soil is of a highly compressible nature.
- The footing area is more than half the building area, it is economical and safe to use raft foundations covering the whole area.
- The use of raft foundation minimizes differential settlement.

  The weight of the soil excavated is almost equal to that of the raft.
- The raft consists of reinforced concrete beams with a relatively thin slab underneath.
- The hydrostatic uplift is also counteracted by the presence of raft foundations.


Main Beam
Secondary
Beam
Columns
Slab
Secondary Beam
Columns
Main Beam

Deep Foundation

In such types of footings, the depth is very large compared to its width.

classified as :

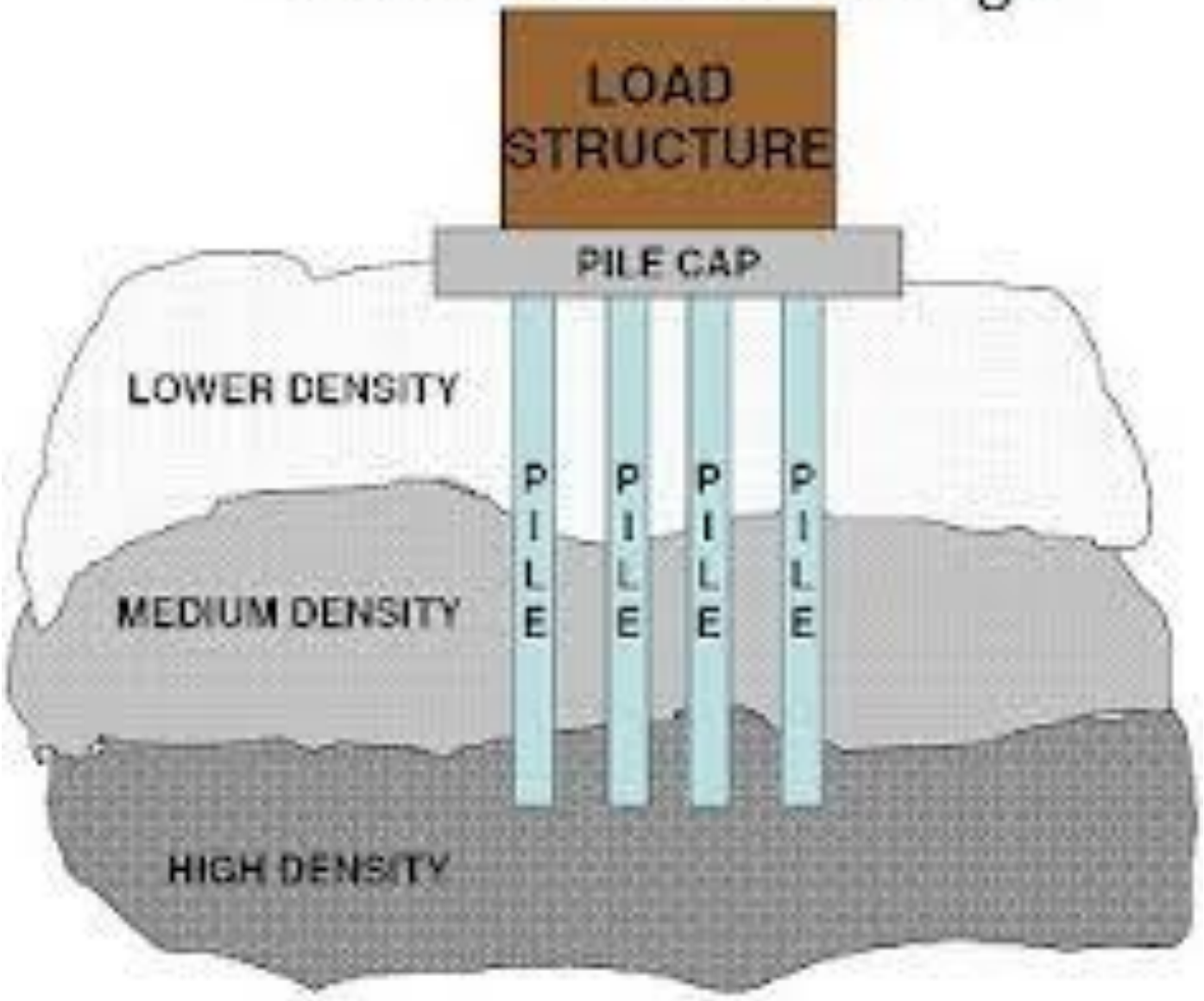
1. pile foundations
2. piers and
3. well foundations.

1.Pile Foundations

Piles are used under the following situations.

- The live load and dead load coming from the structure is very heavy and the distribution of load on soil is uneven.
- The Construction of grillage or raft foundation is not economical.
- The subsoil water level fluctuates appreciably.
- The structure is situated on sea-shore or river bed and there is scouring action of water (Marine Structures)
- In places where canals, deep servers are to be constructed in future.

Pile Foundation Design



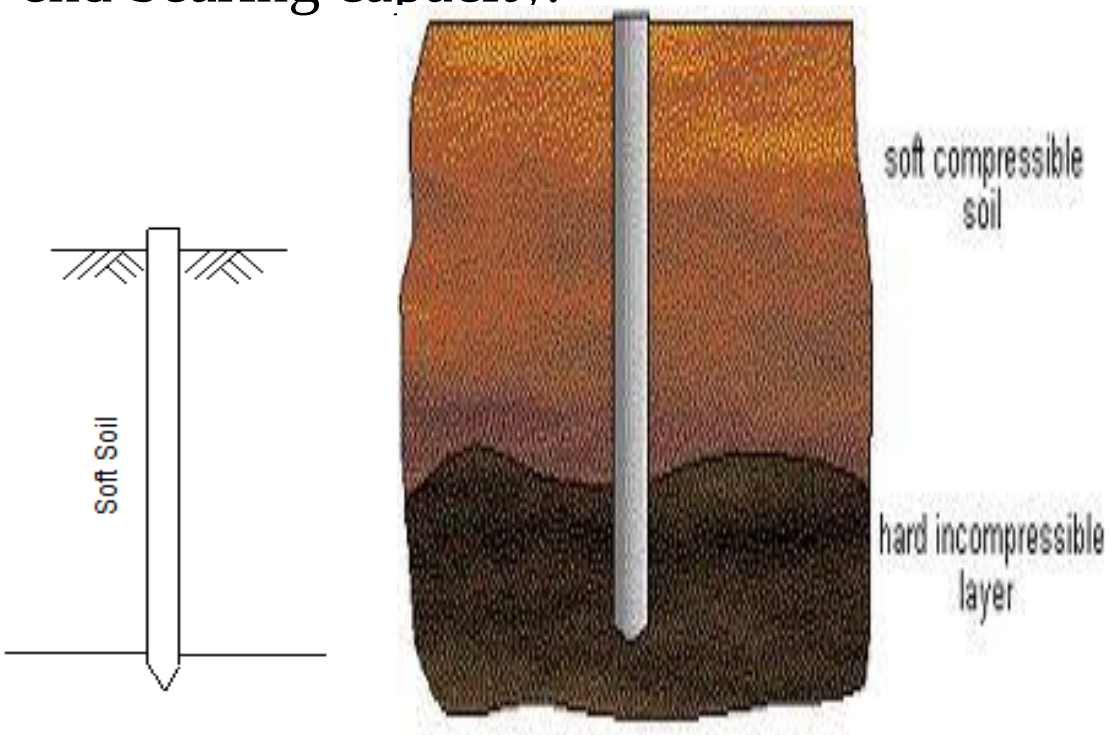
Classification of piles

- Piles are classified based on function as below
  1. End bearing piles
  2. Friction piles
  3. Compaction piles
  4. Sheet pile
  5. Batter pile
  6. Fender pile
  7. Under reamed pile

1.End bearing piles

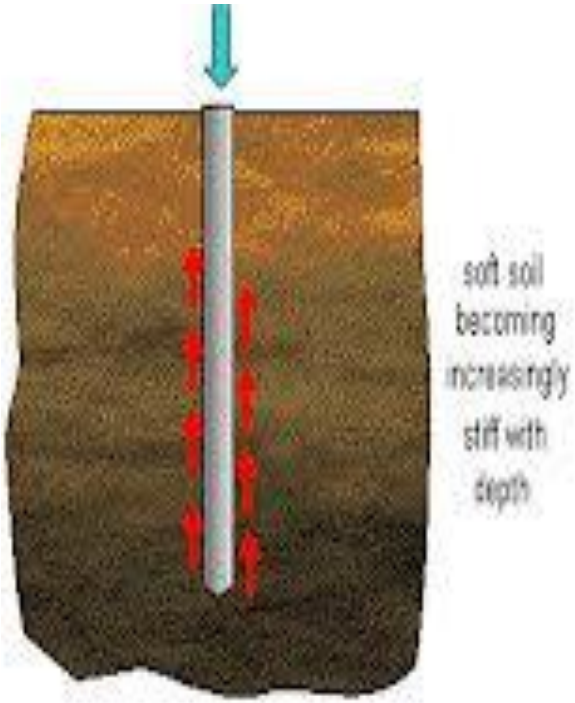
- They are penetrated through soft soil and rest on a hard bed.

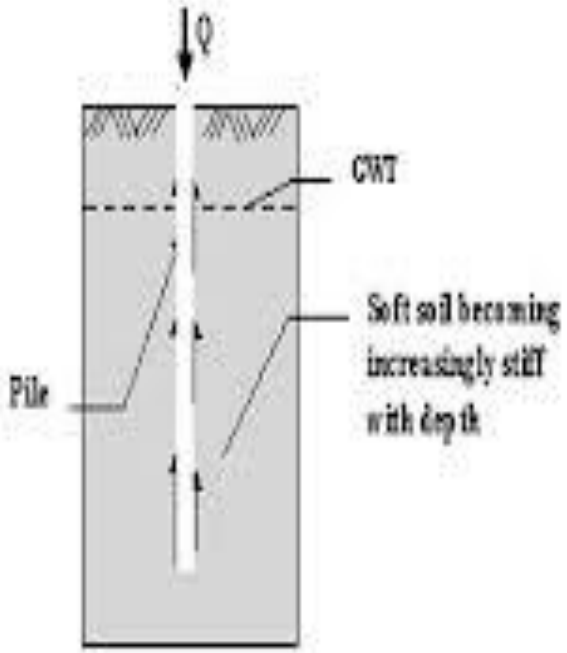
- The friction component from the sides of the pile is less as compared to the end bearing capacity.

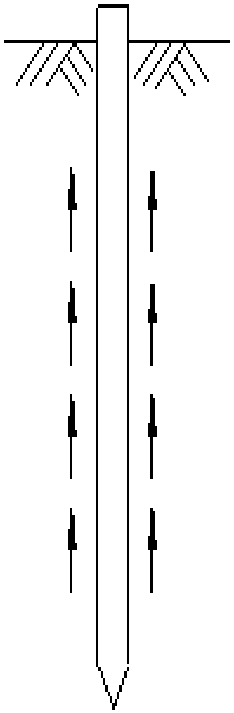


2. Friction piles

- In this type of piles, the frictional resistance offered by the soil on the sides of the pile counteracts the load acting on the pile.
- The load from the superstructure is transmitted through the pile to the soil on the sides.

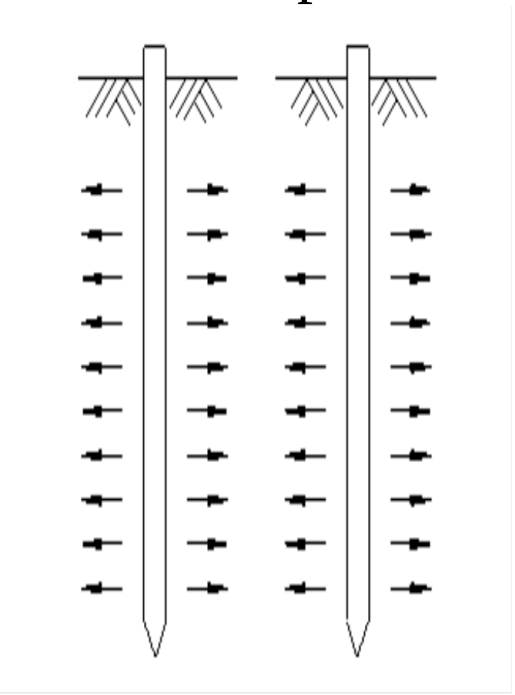


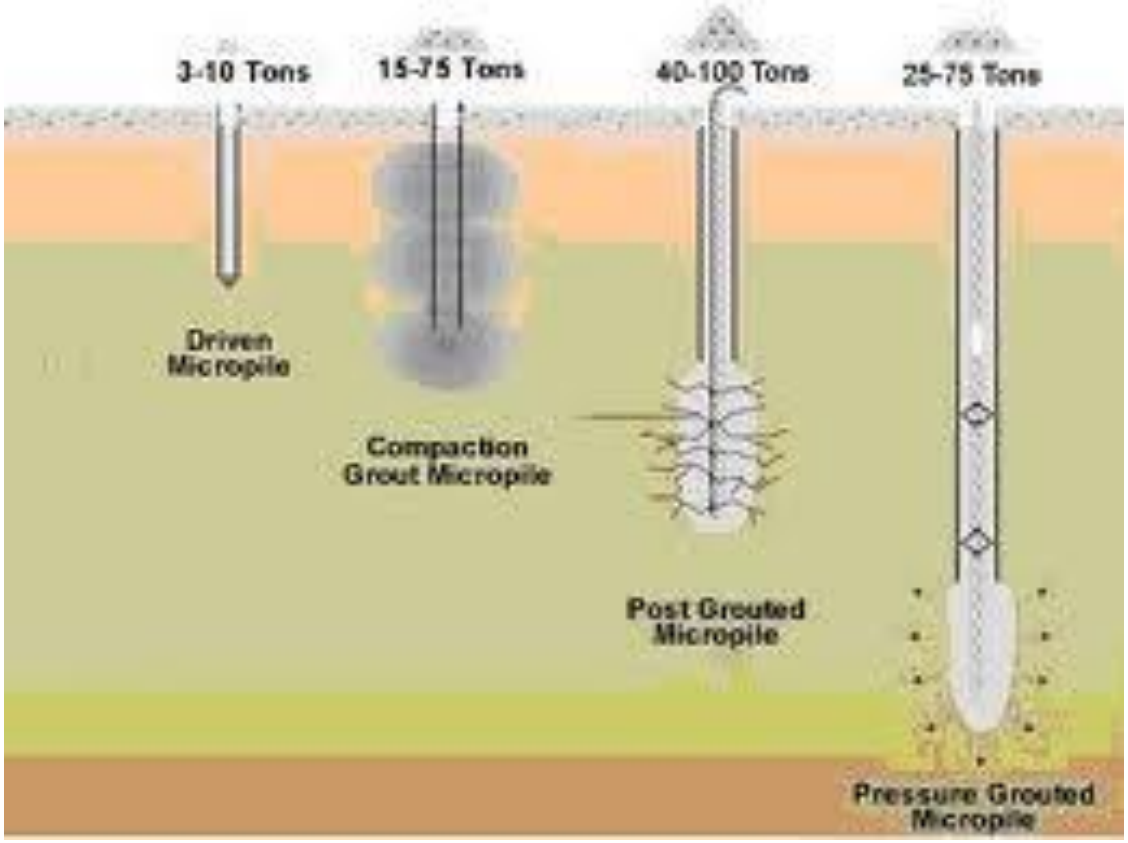




3. Compaction piles

- These piles are used to densify loose granular soils.
- As the piles are driven, the soil-surrounding item gets compacted.





4. Sheet piles

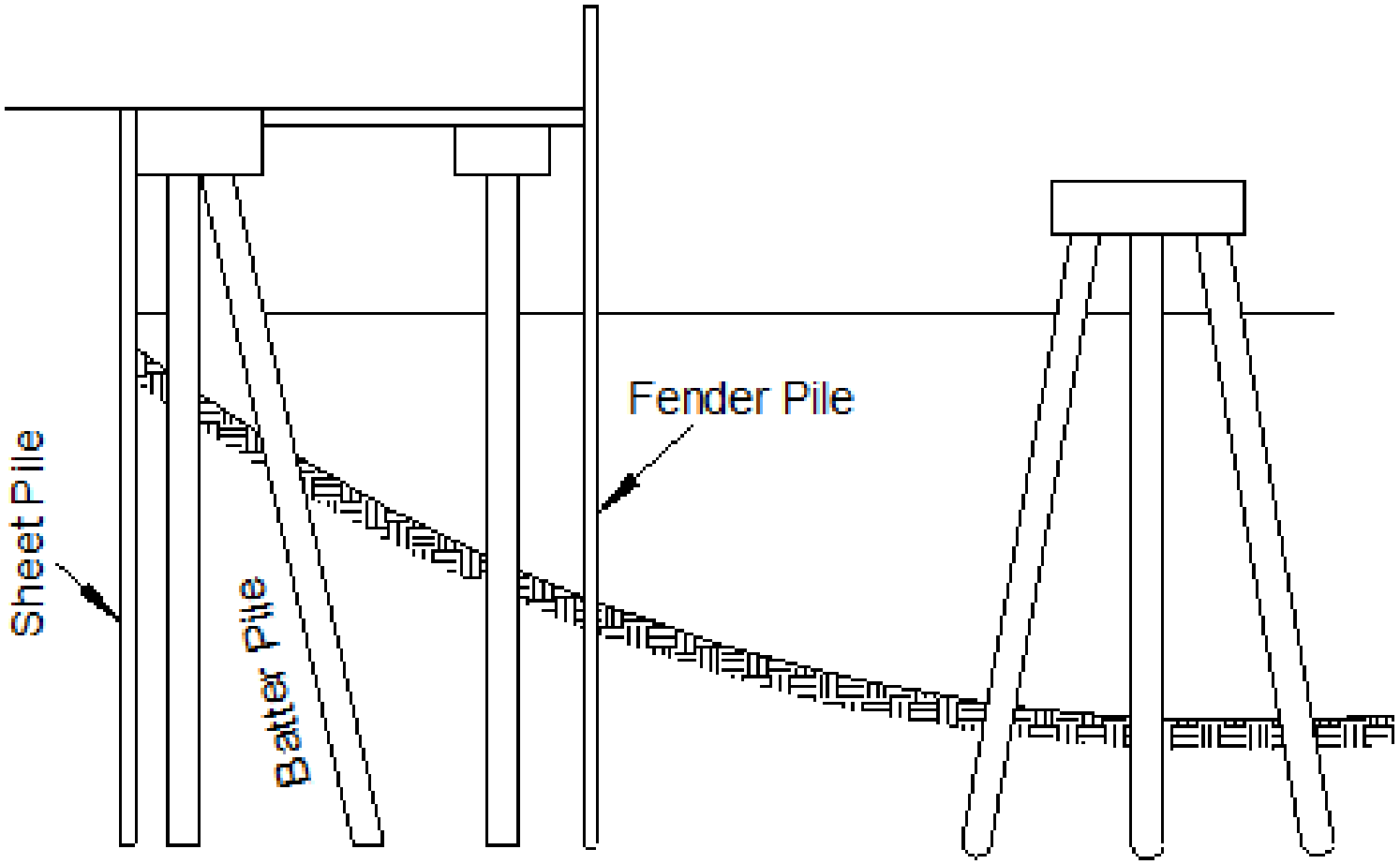
- These piles are thin members of steel or timber used in waterfront structures.
- They can reduce uplift under hydraulic structures.

5. Batter piles

- These are inclined piles existing lateral loads.

6. Fender piles

- These protect the structures against impact from ships and other floating objects.


Sheet Pile
Batter Pile
Fender Pile



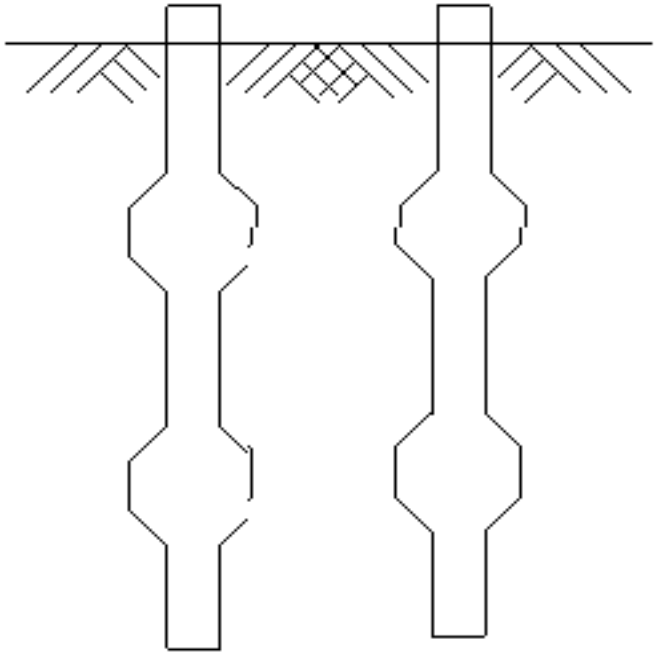
Batter pile

Fender pile



7. Under reamed piles

- The stem of such piles has one or more bulbs known as under ream which increase the bearing capacity of the pile and also resist uplift.



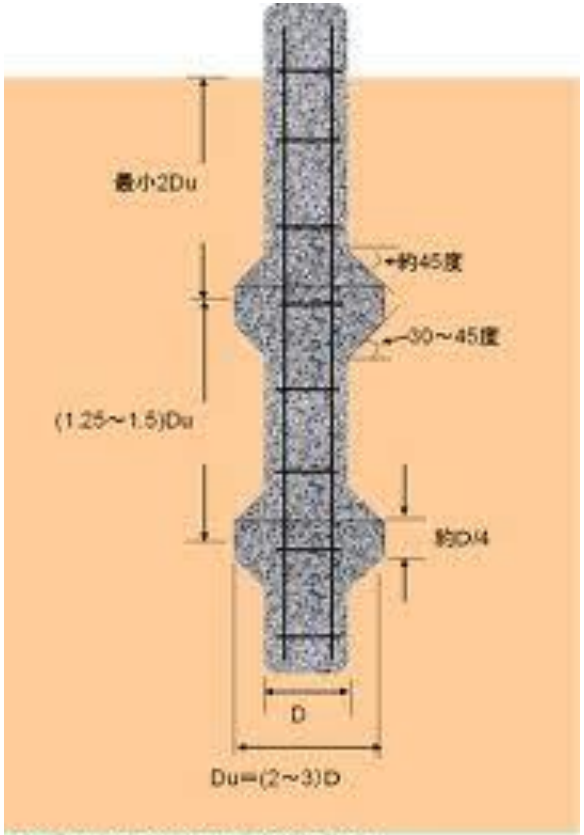
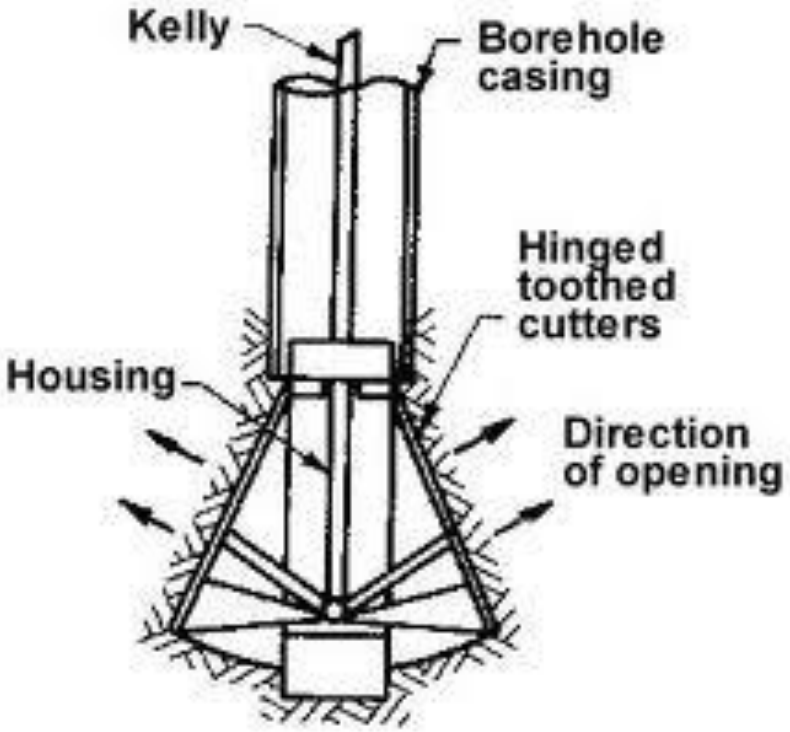


図 翼付き場所打ち杭(インド建築基準)
National Building Code of India 1963


Kelly
Borehole casing
Hinged toothed cutters
Housing
Direction of opening



Classification based on the materials and composition:

1. Cement concrete piles
2. Pre-cast concrete piles
3. Cast in-situ concrete piles
4. Timber piles
5. Steel piles
6. Sand piles
7. Composite piles

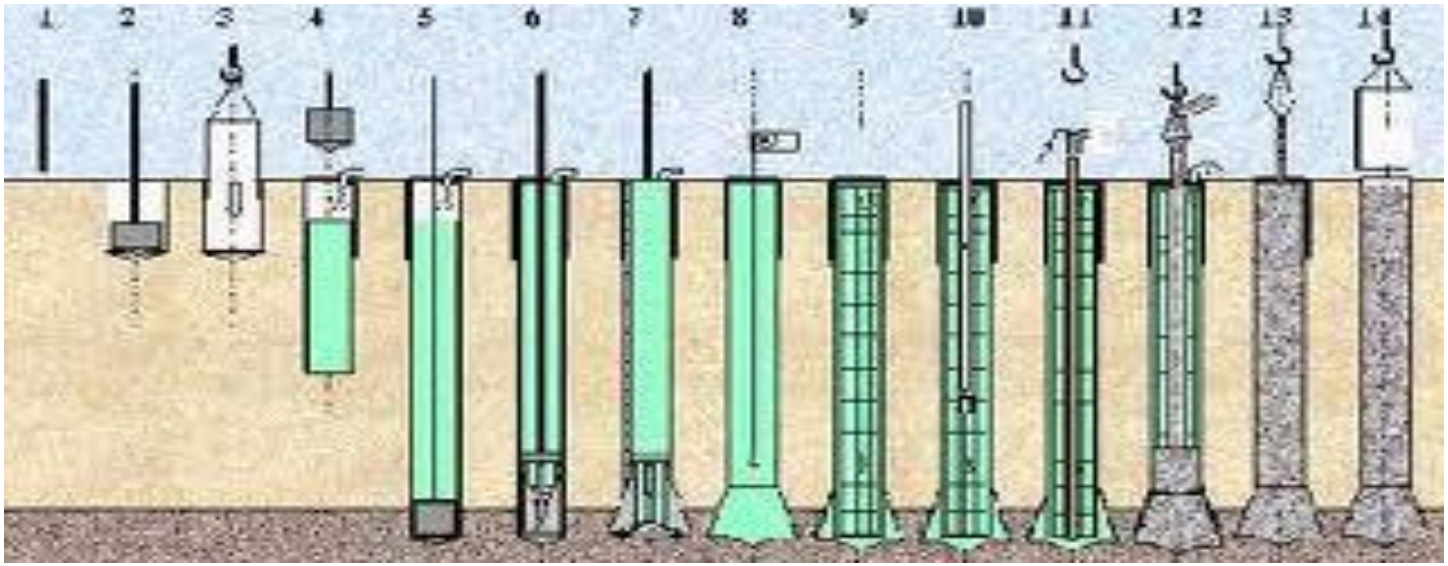
1. Cement concrete piles

- Cement concrete piles possess excellent compressive strength.
- These piles can be reinforced or prestressed.
- They can be divided into following two groups- pre-cast concrete pile and cast in-situ concrete piles

2. Pre-cast concrete piles

- These piles are manufactured in factory.
- The precast concrete piles are generally used for a maximum design load of about 80 tons.
- They may be reinforced to withstand handling stresses.




1
2
3
4
5
6
7
8
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10
11
12
13
14



3. Cast in-situ concrete piles

- In this type a bore is dug into the ground by inserting a casing,.
- This bore is then filled with cement concrete after placing reinforcement, if any. They may be either cased cast in-situ concrete piles or encased cast in-situ concrete piles depending upon whether the casing is kept in position or is withdrawn.

4. Timber piles

- Timber piles are prepared from trunks of trees.
- They may be circular or square. They are 30 to 50 cm in diameter with a length not exceeding 20 times its top width. At the bottom, a cast – iron shoe is provided and at the top, a steel plate is fixed.
- They have small bearing capacity, and are not permanent unless treated. It is very difficult or even impossible to drive these piles into hard stratum.



5. Steel piles

- Steel piles are used in the following three different forms.
- H-Piles , Box Piles and Tube Piles

6. Sand piles

- These piles are formed by making holes in the ground and then filling them by sand.
- The top of the sand piles is filled with concrete to prevent the sand to come upwards due to lateral pressure.

7. Composite piles

- A composite pile is formed when it is a combination either of a bored pile and a driven pile or of driven piles of two different materials.
- They are economical and easy to construct.

